

Machine Learning Basics

Learning Objectives

- Explain the fundamental concept of learning from data
- Distinguish between supervised, unsupervised, and reinforcement learning
- Analyze real-world datasets to identify learning task types

Engage (5-7 minutes)

Show students two photos: one of a cat and one of a dog. Ask: how do you know which is which? You learned from examples as a child. Now imagine teaching a computer the same way—showing it thousands of labeled images. Machine learning is exactly this: systems that improve through experience rather than explicit programming.

Explore (10-12 minutes)

Students receive three datasets: labeled student grades (supervised), unlabeled shopping transactions (unsupervised), and a maze with rewards/penalties (reinforcement). For each, they describe what a machine could learn and how the learning process differs. Groups categorize additional examples and justify their classifications.

Explain (10-12 minutes)

Machine Learning is a subset of AI where systems learn patterns from data without being explicitly programmed. Supervised learning uses labeled input-output pairs to learn a mapping function (e.g., email → spam/not spam). Unsupervised learning finds hidden structure in unlabeled data (e.g., customer segmentation). Reinforcement learning trains agents through trial and error, rewarding desired behaviors (e.g., game-playing agents). Each paradigm suits different problem types and data availability.

Elaborate (8-10 minutes)

Examine the machine learning pipeline: data collection → preprocessing → feature extraction → model training → evaluation → deployment. Discuss the critical role of data quality—“garbage in, garbage out.” Analyze bias in training data and how it produces biased models. Case studies: a fraud detection system (supervised), a recommendation engine (unsupervised), and AlphaGo (reinforcement learning).

Evaluate (5-8 minutes)

Students receive scenario cards and must identify the appropriate ML paradigm, justify their choice, and describe what the training data would look like. Practical exercise: given a simple dataset, identify features, labels, and the learning task. Critical thinking: discuss a case where machine learning fails and why.